School of Computing

M33147-INTELLIGENT DATA AND TEXT ANALYTICS

Coursework #

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Summary Table of Contributions

Group Member	Main Responsibilities	Specific Tasks Completed	Approx. % Contribution
[ID 1] up2541590	Regression, report making and helped in other tasks	Built regression models, analysed results, wrote report sections, and assisted teammates.	30%
[ID 2] up2302123	Clustering, Association Rule Mining and helped in report	Performed clustering, extracted association rules, interpreted results, and refined final report.	30%
[ID 3] up2293411	Data analysis, data preparation for models,Visualization, Classification and helped in report	Ran analyses, created visualizations, trained models, documented findings clearly and finalized document	40%

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Task 1

Introduction to Descriptive Analysis

In this task we will perform the **descriptive analysis** of the **IDTA dataset**. We will explore each column and check the statistical summary and understand the data types and review all the basic information about the dataset. This step helps us to get a clear overview of the data before moving to further analysis.

Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2054 entries, 0 to 2053
Data columns (total 11 columns):
 #   Column                                 Non-Null Count  Dtype
---  ------                                 --------------  -----
 0   Gender                                 2054 non-null   object
 1   Age                                    2054 non-null   int64
 2   Work Pressure                          2054 non-null   int64
 3   Job Satisfaction                       2054 non-null   int64
 4   Sleep Duration                         2054 non-null   object
 5   Dietary Habits                         2054 non-null   object
 6   Have you ever had suicidal thoughts ?  2054 non-null   object
 7   Work Hours                             2054 non-null   int64
 8   Financial Stress                       2054 non-null   int64
 9   Family History of Mental Illness       2054 non-null   object
 10  Depression                             2054 non-null   object
dtypes: int64(5), object(6)
memory usage: 176.6+ KB
```

The dataset contains 2054 entries and 11 columns. Five are numeric (Age, Work Pressure, Job Satisfaction, Work Hours, Financial Stress) and six are categorical (Gender, Sleep Duration, Dietary Habits, Suicidal Thoughts, Family Mental Illness, Depression) and no missing values are present.

Missing Values Check

```
Gender                                   0
Age                                      0
Work Pressure                            0
Job Satisfaction                         0
Sleep Duration                           0
Dietary Habits                           0
Have you ever had suicidal thoughts ?    0
Work Hours                               0
Financial Stress                         0
Family History of Mental Illness         0
Depression                               0
dtype: int64
```

All columns show zero missing values meaning the dataset has no null data. Therefore no data cleaning is required for handling missing entries.

Numerical and Categorical Columns

```
Numerical columns:
Index(['Age', 'Work Pressure', 'Job Satisfaction', 'Work Hours',
       'Financial Stress'],
      dtype='object')
-------------------
Categorical columns:
Index(['Gender', 'Sleep Duration', 'Dietary Habits',
       'Have you ever had suicidal thoughts ?',
       'Family History of Mental Illness', 'Depression'],
      dtype='object')
```

“The dataset contains five numerical columns (Age,Work Pressure, Job Satisfaction, Work Hours, Financial Stress) and six categorical columns (Gender,Sleep Duration, Dietary Habits, Suicidal Thoughts, Family Mental Illness, Depression) helping us understand the data types clearly.”

Numerical Column Summary

	Age	Work Pressure	Job Satisfaction	Work Hours	Financial Stress
count	2054.000000	2054.000000	2054.000000	2054.000000	2054.000000
mean	42.171860	3.021908	3.015093	5.930867	2.978578
std	11.461202	1.417312	1.418432	3.773945	1.413362
min	18.000000	1.000000	1.000000	0.000000	1.000000
25%	35.000000	2.000000	2.000000	3.000000	2.000000
50%	43.000000	3.000000	3.000000	6.000000	3.000000
75%	51.750000	4.000000	4.000000	9.000000	4.000000
max	60.000000	5.000000	5.000000	12.000000	5.000000

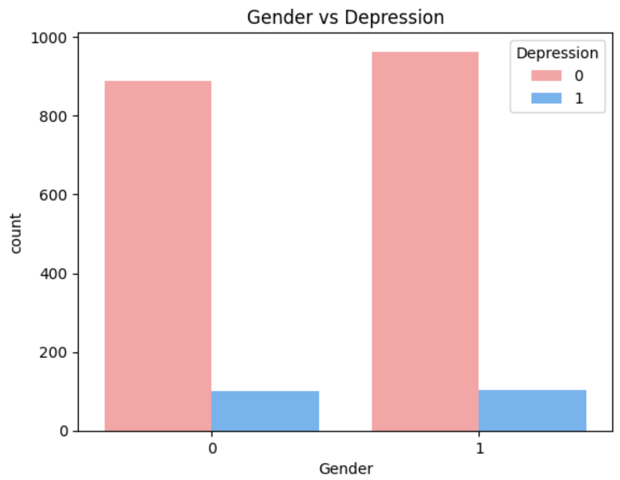
The statistics show unique ranges for each numeric column that Age spans 18–60, Work Pressure and Job Satisfaction range 1–5, Work Hours vary widely from 0–12, and Financial Stress ranges 1–5, indicating different scales and variability.

Categorical Value Distribution

```
Gender
1    1066
0     988
Name: count, dtype: int64
________________________________________
Sleep Duration
7.5    530
4.0    525
5.5    505
9.0    494
Name: count, dtype: int64
________________________________________
Dietary Habits
0    713
2    681
1    660
Name: count, dtype: int64
________________________________________
Have you ever had suicidal thoughts ?
0    1065
1     989
Name: count, dtype: int64
________________________________________
Family History of Mental Illness
0    1046
1    1008
Name: count, dtype: int64
________________________________________
Depression
0    1851
1     203
Name: count, dtype: int64
________________________________________
```

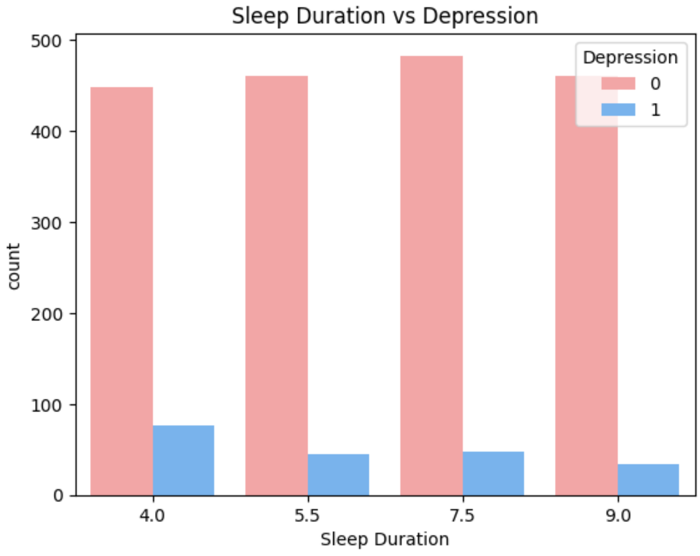
Categorical columns show balanced distributions: gender is nearly equal and sleep duration categories are similar, dietary habits are evenly spread and suicidal thoughts and family mental illness are almost 50–50 while depression is highly imbalanced with only 203 positive cases.

Visualization: Gender vs Depression



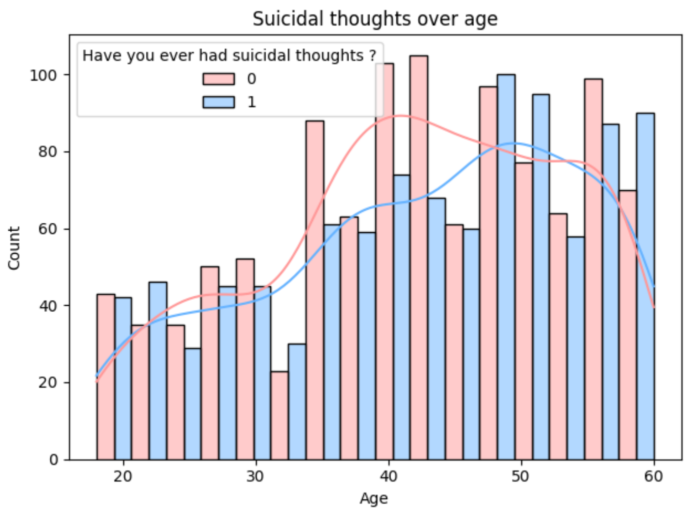
The Gender vs Depression chart show that females (0) include around 880 non-depressed and about 100 depressed individuals, while males (1) show roughly 960 non depressed and around 100 depressed cases. Since both genders have nearly identical depression counts, gender does not significantly influence depression levels in this dataset.

Visualization: Sleep Duration vs Depression



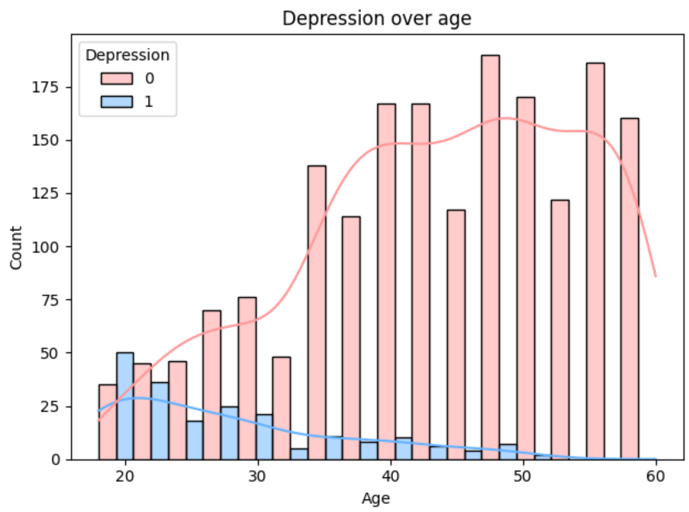
The Sleep Duration vs Depression chart shows that all sleep groups have high non depressed counts about 450–480 individuals. Depression case vary slightly around 80 for 4 hours, 45–55 for 5.5 and 7.5 hours and about 35 for 9 hours. Shorter sleep (4 hours) shows the highest depression count suggesting inadequate sleep may relate to higher depression levels.

Visualization: Suicidal Thoughts Over Age – Interpretation



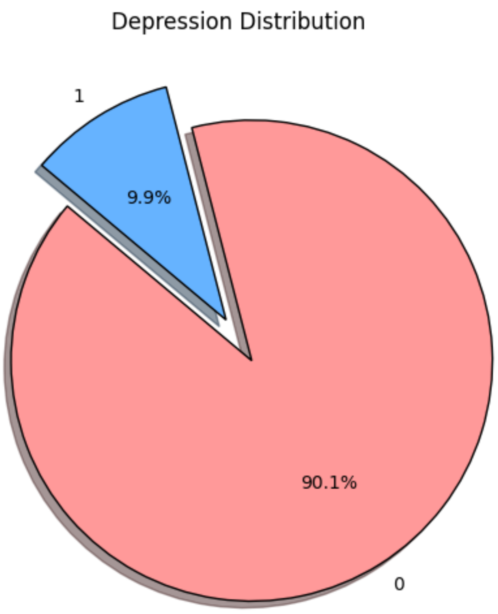
The chart shows how suicidal thoughts vary across age groups. Individuals aged 35–50 consistently show higher counts for both ‘No’ and ‘Yes’ responses. Suicidal thoughts increase notice from the early 30s peaking around ages 40–50. Younger (18–25) and older (55–60) groups show lower counts overall. This suggests middle aged adults report suicidal thoughts more frequently in this dataset.

Visualization: Depression Over Age – Interpretation



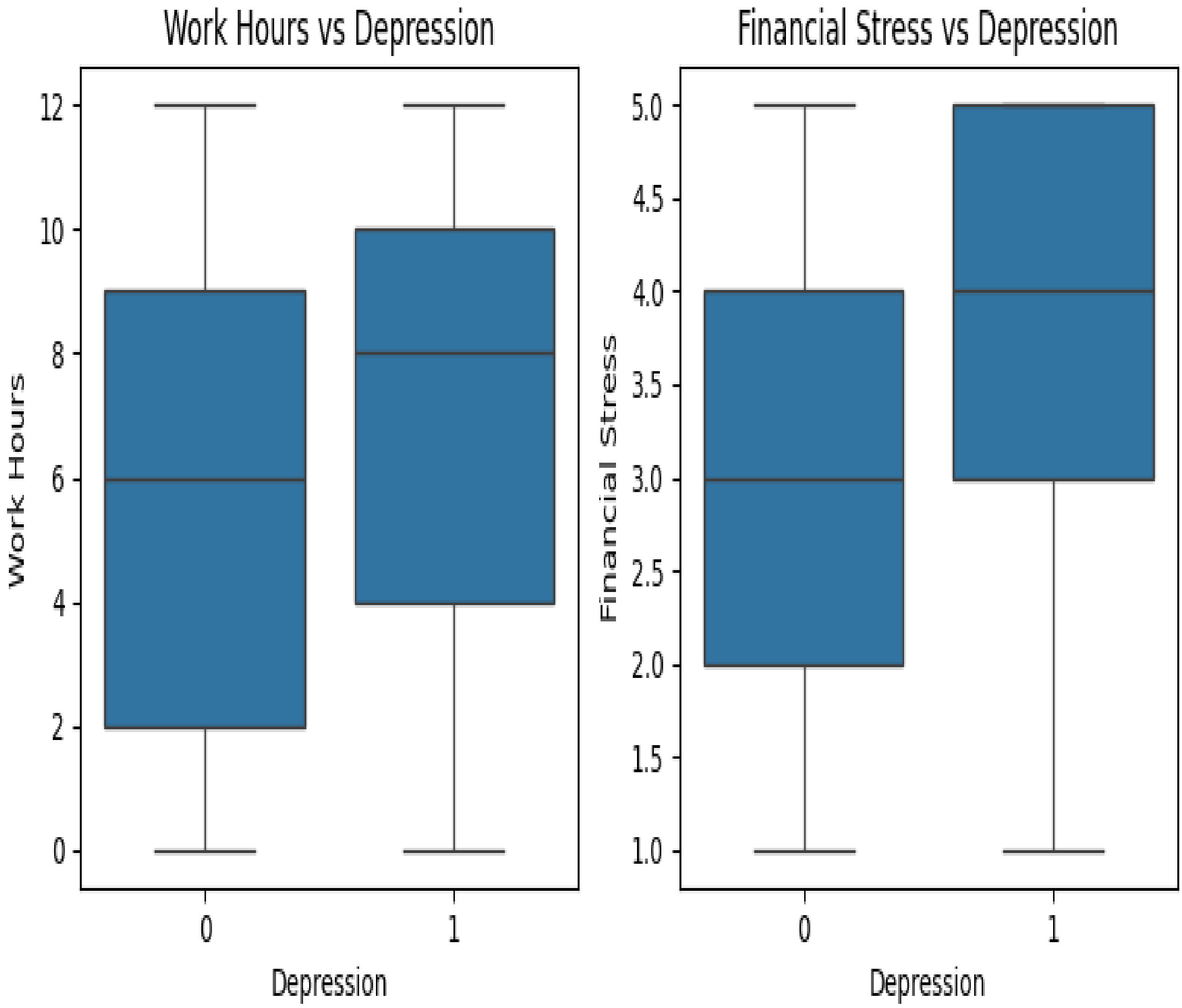
The Depression over age chart shows that non depressed individuals increase steadily from age 25 to 50, reaching counts above 150–190. In contrast depression cases decrease with age around 50 cases at age 20, dropping sharply after 30 and approaching zero by age 50. This suggests depression is more common among younger adults in this dataset.

Visualization: Depression Distribution – Interpretation



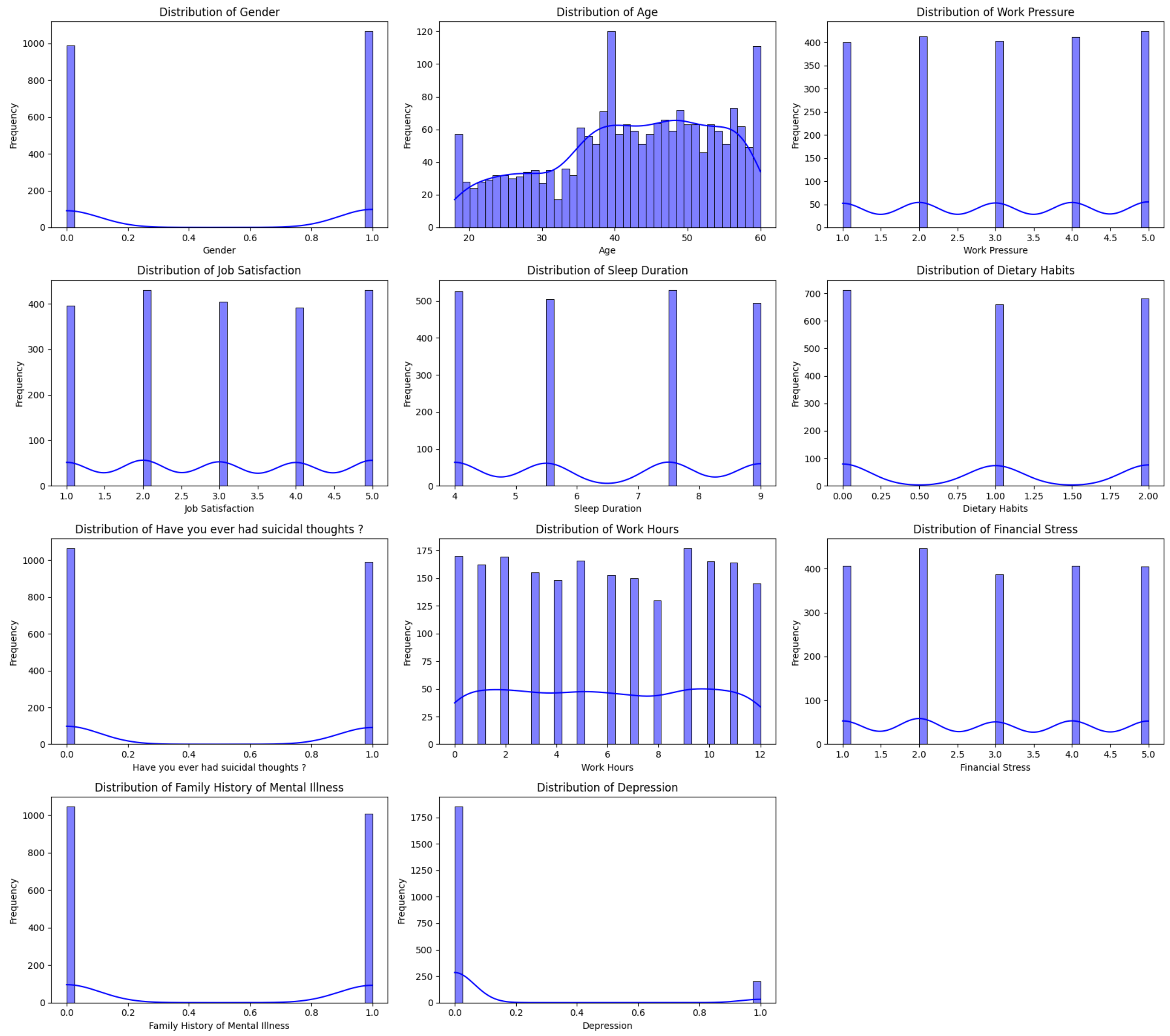
The pie chart shows a severe class imbalance about 90% are not depressed and only 10% are depressed. Such imbalance can cause classification models to become biased toward predict the majority class. Therefore, Random Oversampling was applied to increase the depressed cases ensuring balanced classes and improving model training, accuracy, and fairness.

Visualization: Work Hours & Financial Stress vs Depression



The Work Hours boxplot shows that depressed individuals tend to work slightly longer hours (median ≈ 8) compared to non depressed individuals (median ≈ 6). The Financial Stress boxplot shows a clearer pattern depressed individuals have higher stress levels (median ≈ 4–5) than non-depressed individuals (median ≈ 3). This suggests financial stress is more strongly associated with depression than working hours.

Visualization: Distribution of All Features – Interpretation



Key Observations

- Most features (Gender, Work Pressure, Job Satisfaction,Sleep Duration, Dietary Habits, Suicidal Thoughts, Family History) are **evenly distributed** with no major outliers.
- Age ranges between **18–60** mostly in the **30–55** group.
- Work Hours vary widely from **0–12**.
- **Depression is highly imbalanced** (1851 No, 203 Yes).

Conclusion

Most features show normal distribution without outliers. The only major issue is the **imbalanced Depression variable** which required **Random Oversampling** to prepare the data for a classification models.

TASK 2

Classification

Introduction

In this task we built classification model to predict whether a person is depressed or not. Since the dataset was highly imbalanced that we applied Random Oversampling to balance the classes. After preparing the data we trained the Logistic Regression, SVM and Decision Tree models and evaluated their performance using accuracy, precision, recall, and F1-scores.

Categorical Data Conversion

In this step we converted all categorical values into numerical form so that the data becomes suitable for classification modeling.

Random Oversampling (Handling Class Imbalance)

In this step we used the *RandomOverSampler* method to balance the dataset. Since depression cases were very low compared to non depressed cases, oversampling was applied to increase the minority class making the data suitable for building a fair and accurate classification model.

```
Depression
1    1481
0    1481
Name: count, dtype: int64
```

Logistic Regression Parameters and Results

We trained a Logistic Regression model using **L2 regularization**, the **liblinear solver**, **C = 0.01** (strong regularization) and **max_iter = 100**. These settings helped control overfitting and work well with binary classification. After applying Random Oversampling the model performed very well, achieving of **95% accuracy**.

Both classes show strong results :

- **NotDepressed :** F1-score=**0.95**
- **Depressed :** F1-score = **0.95**

This means the model can correctly classify depressed and non depressed individual with high reliability.

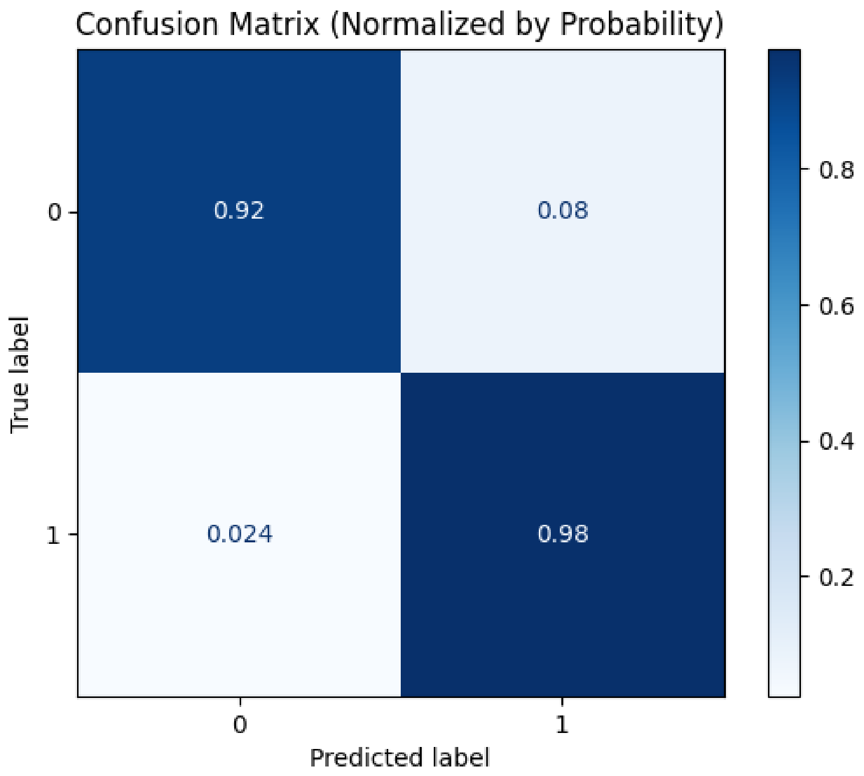
Output

```
              precision    recall  f1-score   support

           0       0.92      0.97      0.95      1399
           1       0.98      0.92      0.95      1563

    accuracy                           0.95      2962
   macro avg       0.95      0.95      0.95      2962
weighted avg       0.95      0.95      0.95      2962
```

The confusion matrix



SVM Hyperparameter Tuning (GridSearchCV)

To improve a SVM performance we used **GridSearchCV** to find the best parameters. First we tuned **C** values using an RBF kernel and the best result was:

- Best C = 100

Next we tuned the **gamma** parameter (with C fixed at 100) and GridSearchCV gave:

- Best gamma = 1

These optimal parameters help the SVM model fit the data better and improve classification accuracy.

output

```
Best C: {'C': 100}
Best gamma: {'gamma': 1}
```

SVM Model Results (Using Best Parameters)

After tuning the hyperparameters with the GridSearchCV we trained the SVM model using **C = 100** and **gamma = 0.01**. These optimized values helped the model fit to the oversampled data very effectively.

Performance Results

The SVM achieved:

- **Precision = 1.00**
- **Recall = 1.00**
- **F1-score = 1.00**
- **Accuracy = 100%**

Both classes (0 and 1) have perfect scores meaning the model correctly classified every instance in the resampled training data.

Conclusion

The SVM model performed **exceptionally well** on the balanced dataset showing the perfect prediction scores. This indicates an excellent fit likely due to the combination of oversampling and optimized hyperparameters.

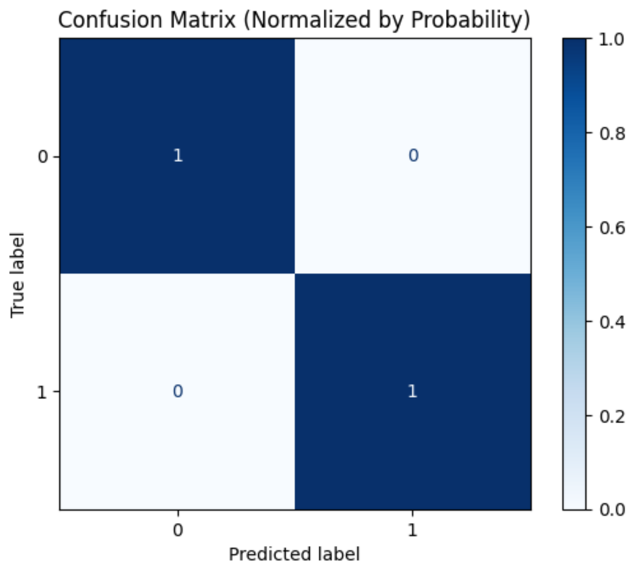
Output

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      1481
           1       1.00      1.00      1.00      1481

    accuracy                           1.00      2962
   macro avg       1.00      1.00      1.00      2962
weighted avg       1.00      1.00      1.00      2962
```

The confusion matrix:



Decision Tree Classifier Results

The Decision Tree model was trained using the Gini criterion with no maximum depth that allowing the tree to fully learn patterns from the oversampled dataset . After training the model achieved perfect evaluation scores.

Performance Results

- **Precision :** 1.00 for both classes
- **Recall :** 1.00 for both classes
- **F1-score :** 1.00 for both classes
- **Accuracy :** 100%

Conclusion

The Decision Tree model perfectly classified all samples in the balanced dataset. Since the dataset was oversampled and the tree had no depth limit so that it was able to learn all patterns clearly resulting in a **perfect prediction performance**.

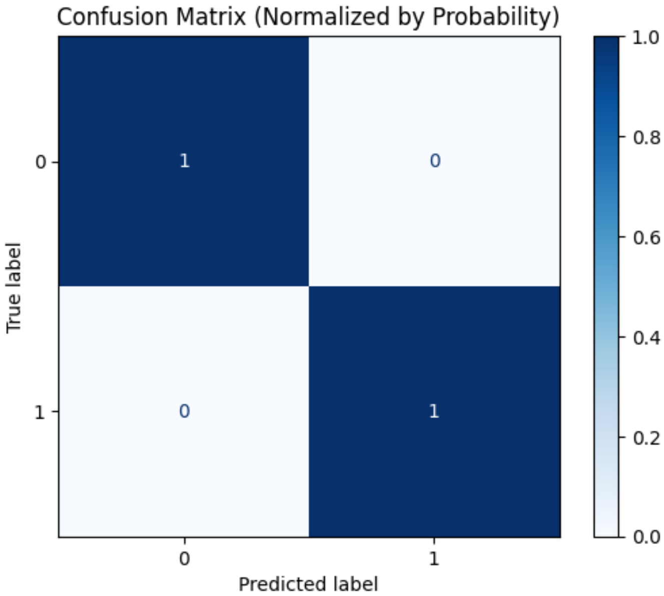
Output

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00      1481
           1       1.00      1.00      1.00      1481

    accuracy                           1.00      2962
   macro avg       1.00      1.00      1.00      2962
weighted avg       1.00      1.00      1.00      2962
```

The confusion matrix:



Conclusion

All three models performed extremely well after the oversampling. Logistic Regression achieved 95% accuracy while both SVM and Decision Tree reached perfect 100% scores on the resampled data. The oversampling process greatly improved model performance by balancing the classes allowing each classifier to learn depression patterns effectively and make accurate predictions.

TASK 3

Regression

Introduction

In this task we applied different regression models to predict a continuous target variable using our processed dataset. Before modeling all categorical columns were label encoded and numerical features were normalized using MinMaxScaler. We then trained Linear Regression, Ridge Regression and Random Forest Regression to evaluate how well each model could learn and predict the target values. Model performance was measured using MAE, MSE, RMSE, and R^2 score.

Label Encoding for Regression Preparation

In this step we converted all categorical columns into a numeric values using the Label Encoder method. Columns such as Gender, Sleep Duration, Dietary Habits, Suicidal Thoughts, Family Mental Illness and Depression were transformed so the dataset becomes fully numeric and ready for the further regression processing.

Output

```
   Gender  Age  Work Pressure  Job Satisfaction  Sleep Duration  \
0       0   37              2                 4               1
1       1   60              4                 3               0
2       0   42              2                 3               0
3       0   44              3                 5               1
4       1   48              4                 3               1

   Dietary Habits  Have you ever had suicidal thoughts ?  Work Hours  \
0               1                                      0           6
1               2                                      1           0
2               1                                      0           0
3               0                                      1           1
4               1                                      1           6

   Financial Stress  Family History of Mental Illness  Depression
0                 2                                 0           0
1                 4                                 1           0
2                 2                                 0           0
3                 2                                 1           0
4                 5                                 1           0
```

Normalization Using MinMaxScaler

In this step we normalized all the numerical columns using the MinMaxScaler method. This scales every numeric value into a range between 0 and 1 and ensuring that all the features have equal weight and improving the performance of regression models.

Output

```
   Gender       Age  Work Pressure  Job Satisfaction  Sleep Duration  \
0     0.0  0.452381           0.25              0.75        0.333333
1     1.0  1.000000           0.75              0.50        0.000000
2     0.0  0.571429           0.25              0.50        0.000000
3     0.0  0.619048           0.50              1.00        0.333333
4     1.0  0.714286           0.75              0.50        0.333333

   Dietary Habits  Have you ever had suicidal thoughts ?  Work Hours  \
0             0.5                                    0.0    0.500000
1             1.0                                    1.0    0.000000
2             0.5                                    0.0    0.000000
3             0.0                                    1.0    0.083333
4             0.5                                    1.0    0.500000

   Financial Stress  Family History of Mental Illness  Depression
0              0.25                               0.0         0.0
1              0.75                               1.0         0.0
2              0.25                               0.0         0.0
3              0.25                               1.0         0.0
4              1.00                               1.0         0.0
```

Linear Regression Results

We trained a Linear Regression model on the processed dataset and evaluated its performance using a common regression metrics.

The results are :

- **MAE :** 0.19
- **MSE :** 0.057
- **RMSE :** 0.23
- **R^2 Score :** 0.25

Interpretation

The error values(MAE, MSE, RMSE) are relatively low, meaning predictions are reasonably close to the actual values. However, the **R^2 score is 0.25**, which indicates the model explains only **25% of the variance** in the target variable. This means Linear Regression does not capture the relationship strongly and may not be the best model for this dataset.

Output

```
--- Linear Regression Results ---
MAE: 0.19805398427065385
MSE: 0.05735487762862227
RMSE: 0.239488783930735
R² Score: 0.25292105805814413
```

Ridge Regression Results

We applied Ridge Regression with α = 1.0 which adds L2 regularization to reduce overfitting and stabilize the model. After training and testing the model we obtained the following results:

- **MAE :** 0.1982
- **MSE :** 0.0574
- **RMSE :** 0.2396
- **R^2 Score :** 0.2524

Interpretation

The error values (MAE, MSE, RMSE) are very similar to Linear Regression and the **R^2 score remains around 0.25** meaning the model explains about **25% of the variance** in the target variable. This indicates that adding regularization did not significantly improve the model performance.

Coefficients

The Ridge coefficients shows how much the each feature contributes to the prediction with both positive and negative values indicating varying influence. Regularization forces the values to stay smaller and helping prevent overfitting.

Output

```
--- Ridge Regression Results ---
MAE: 0.19821739092895124
MSE: 0.05739519378939429
RMSE: 0.23957294043650734
R² Score: 0.25239591780890946
Ridge Coefficients: [ 0.00202998  0.05518236 -0.09059432 -0.01834053  0.02139944  0.07475894
  0.05275923  0.07086726  0.02005306 -0.4698093 ]
```

Random Forest Regression Results

We trained a Random Forest Regressor with **200 trees(n_estimators = 200)** and fixed random state for the consistency. After fitting the model on the training data we tested it using the regression metrics below:

- **MAE:** 0.2069
- **MSE:** 0.0641
- **RMSE:** 0.2533
- **R^2 Score:** 0.1650

Interpretation

The error values (MAE, MSE, RMSE) are slightly higher compared to Linear and Ridge Regression. The **R^2 score is 0.16** meaning that the model explains only **16% of the variance** in the target variable lower than the previous models.

This indicates that Random Forest did **not perform well** for this prediction task possibly due to limited signal in the data or noise affecting tree based learning.

Output

```
--- Random Forest Regression Results ---
MAE: 0.2069370210371253
MSE: 0.06410096176551276
RMSE: 0.25318167738901004
R² Score: 0.16504958822653892
```

Conclusion

Among the regression models tested, none achieved strong predictive performance. Linear Regression and Ridge Regression showed similar results with an R^2 score around 0.25 meaning they explained about 25% of the variance. Random Forest performed even lower with an R^2 of 0.16. While error values remained low and the overall variance explained was limited indicating that the available features do not strongly predict the selected target variable.

TASK 4

Association Rule Mining

Introduction

In this task we applied Association Rule Mining to discover hidden relationships between different categorical features in the dataset. After converting the data into one-hot encoded format then we used the Apriori algorithm to generate frequent itemsets and then created association rules based on the confidence and lift. This helped us identify the patterns that commonly occur together in the data.

One-Hot Encoding and Frequent Itemset Generation

In this step we converted all categorical data into one-hot encoded format using pd.get_dummies(), creating separate binary columns for the each category. After encoding we applied the Apriori algorithm with a minimum support of 3% to generate the frequent itemsets. This gives us combinations of items that appear frequently in the dataset.

Output

```
Number of frequent itemsets: 575
    support                          itemsets
0  0.481013                   (Gender_Female)
1  0.518987                     (Gender_Male)
2  0.245862        (Sleep Duration_5-6 hours)
3  0.258033        (Sleep Duration_7-8 hours)
4  0.255599  (Sleep Duration_Less than 5 hours)
```

Generating Association Rules Using Apriori

In this step we applied the association_rules() function on the frequent itemsets obtained earlier. We used "confidence" as the evaluation metric with a minimum threshold of 0.3. This means only rules with at least 30% confidence were selected. The output shows the total number of rules generated and displays the first few rules along with their metrics such as support,confidence, lift, leverage, conviction and others.

Output

```
Total rules generated: 2017
                         antecedents              consequents  \
0         (Sleep Duration_5-6 hours)          (Gender_Female)
1         (Sleep Duration_7-8 hours)          (Gender_Female)
2  (Sleep Duration_Less than 5 hours)          (Gender_Female)
3  (Sleep Duration_More than 8 hours)          (Gender_Female)
4                    (Gender_Female)  (Dietary Habits_Healthy)

   antecedent support  consequent support   support  confidence      lift  \
0            0.245862            0.481013  0.121227    0.493069  1.025065
1            0.258033            0.481013  0.116845    0.452830  0.941410
2            0.255599            0.481013  0.128530    0.502857  1.045414
3            0.240506            0.481013  0.114411    0.475709  0.988973
4            0.481013            0.331548  0.162123    0.337045  1.016578

   representativity  leverage  conviction  zhangs_metric   jaccard  certainty  \
0               1.0  0.002964    1.023784       0.032424  0.200161   0.023231
1               1.0 -0.007272    0.948494      -0.077389  0.187793  -0.054303
2               1.0  0.005583    1.043940       0.058357  0.211369   0.042091
3               1.0 -0.001276    0.989883      -0.014468  0.188452  -0.010220
4               1.0  0.002644    1.008291       0.031422  0.249251   0.008222

   kulczynski
0    0.372547
1    0.347873
2    0.385032
3    0.356781
4    0.413016
```

Filtering Depression Rules

We checked the association rules to find those that predict **Depression = Yes** but none were found. Since no rules had “Depression_Yes” on the right-hand side, the code displayed the **top 5 strongest rules overall** instead. All these rules predict **Depression_No** with **100% confidence** and a lift of about **1.10**, meaning the dataset shows the stronger patterns related to *not being depressed*.

```
Depression YES column name: Depression_Yes
Rules with Depression=Yes on RHS: 0

⚠ No rules found with Depression=Yes on RHS. Showing top 5 rules overall instead.


Top 5 rules (raw):
                                           antecedents      consequents  \
1594  (Sleep Duration_More than 8 hours, Have you ev...  (Depression_No)
1774  (Family History of Mental Illness_No, Gender_F...  (Depression_No)
793   (Sleep Duration_7-8 hours, Gender_Female, Have...  (Depression_No)
1666  (Family History of Mental Illness_No, Have you...  (Depression_No)
1908  (Gender_Male, Family History of Mental Illness...  (Depression_No)

       support  confidence     lift
1594  0.050146         1.0  1.10967
1774  0.046738         1.0  1.10967
793   0.053554         1.0  1.10967
1666  0.093476         1.0  1.10967
1908  0.031646         1.0  1.10967
```

The 5 rules are:

1. **Rule 1**
   If: Sleep > 8 hours, No suicidal thoughts, Healthy diet
   Then: Depression = No
   Meaning: People with good sleep has no suicidal thoughts and healthy habits are always non-depressed.

2. **Rule 2**
   If: No family mental illness, Female, No suicidal thoughts, Healthy diet
   Then: Depression = No
   Meaning: These combined healthy conditions always lead to non-depressed cases.

3. **Rule 3**
   If: Sleep 7–8 hours, Female, No suicidal thoughts
   Then: Depression = No
   Meaning: Females with good sleep and no suicidal thoughts are always non-depressed.

4. **Rule 4**
   If: No family mental illness, No suicidal thoughts, Healthy diet
   Then: Depression = No
   Meaning: This is the most common rule and always leads to non-depressed cases.

5. **Rule 5**
   If: Male, Family mental illness = Yes, Sleep > 8 hours, No suicidal thoughts
   Then: Depression = No
   Meaning: Even with family history, good sleep and no suicidal thoughts indicate no depression.

Output:

```
IF Sleep Duration_More than 8 hours, Have you ever had suicidal thoughts ?_No, Dietary Habits_Healthy  THEN  Depression_No
  support    = 0.050
  confidence = 1.000
  lift       = 1.110
------------------------------------------------------------------------------------------------------------------------------------
IF Family History of Mental Illness_No, Gender_Female, Have you ever had suicidal thoughts ?_No, Dietary Habits_Healthy  THEN  Depression_No
  support    = 0.047
  confidence = 1.000
  lift       = 1.110
------------------------------------------------------------------------------------------------------------------------------------
IF Sleep Duration_7-8 hours, Gender_Female, Have you ever had suicidal thoughts ?_No  THEN  Depression_No
  support    = 0.054
  confidence = 1.000
  lift       = 1.110
------------------------------------------------------------------------------------------------------------------------------------
IF Family History of Mental Illness_No, Have you ever had suicidal thoughts ?_No, Dietary Habits_Healthy  THEN  Depression_No
  support    = 0.093
  confidence = 1.000
  lift       = 1.110
------------------------------------------------------------------------------------------------------------------------------------
IF Gender_Male, Family History of Mental Illness_Yes, Sleep Duration_More than 8 hours, Have you ever had suicidal thoughts ?_No  THEN  Depression_No
  support    = 0.032
  confidence = 1.000
  lift       = 1.110
------------------------------------------------------------------------------------------------------------------------------------
```

Conclusion

The results showed that strong rules were found mainly for the "Depression_No" group and no rules appeared for "Depression_Yes" due to fewer positive cases. The top rules revealed that good sleep, healthy diet, no suicidal thoughts and no family mental illness were consistently associated with non-depressed individuals. These rules highlight lifestyle patterns strongly linked with lower depression levels in the dataset.

TASK 5

Clustering

Introduction

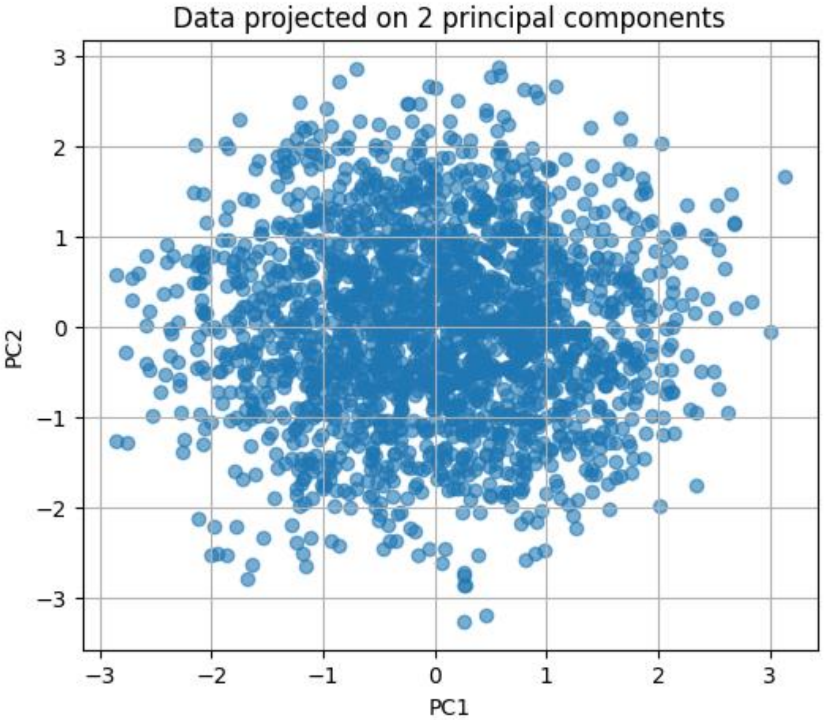
In this task we performed unsupervised learning using clustering methods to find natural groups within the dataset. Since the clustering does not use labels we first prepared the data by encoding categorical values and standardizing all features. Then we applied K-Means and Hierarchical Clustering to check whether the dataset forms meaningful clusters. We also used PCA for visualization and the Elbow Method to determine the best number of clusters.

Clustering Data Preparation

In this step we prepared the dataset for clustering. First, all categorical columns such as Gender, Sleep Duration, Dietary Habits, Suicidal Thoughts, and Family Mental Illness were converted into numeric format using Label Encoding. After encoding that we applied StandardScaler to normalize all features ensuring they are on the same scale before performing clustering.

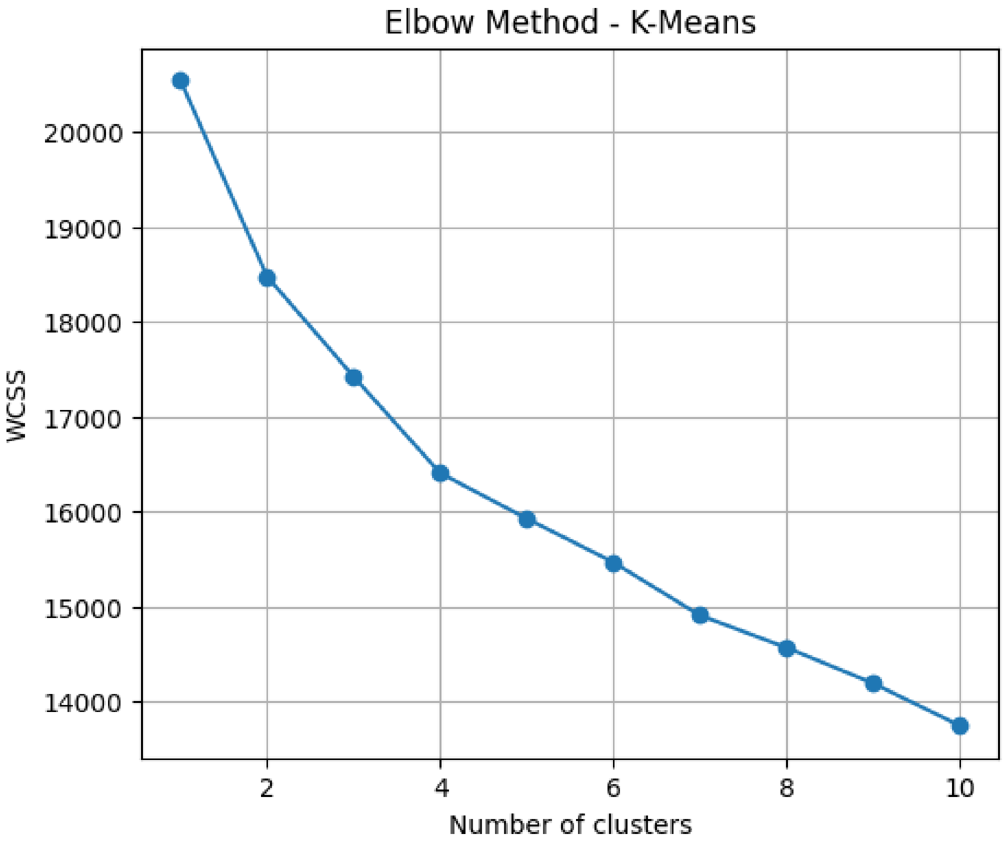
PCA Visualization Before Clustering

In this step we applied Principal Component Analysis (PCA) to reduce the scaled dataset into two principal components. This helps us visualize the overall structure of the data in a 2D space before performing clustering. The scatter plot shows how the points are spread that indicating the general distribution and helping us to understand whether natural cluster patterns exist.



Elbow Method for Choosing Optimal Clusters

In this step we used the Elbow Method to help determine the best number of clusters for K-Means. The WCSS (Within-Cluster Sum of Squares) was calculated for 1 to 10 clusters. The plot shows how WCSS decreases as the number of clusters increases. The curve begins to flatten around 2–4 clusters, indicating that this range is likely the most suitable choice for the K-Means clustering.



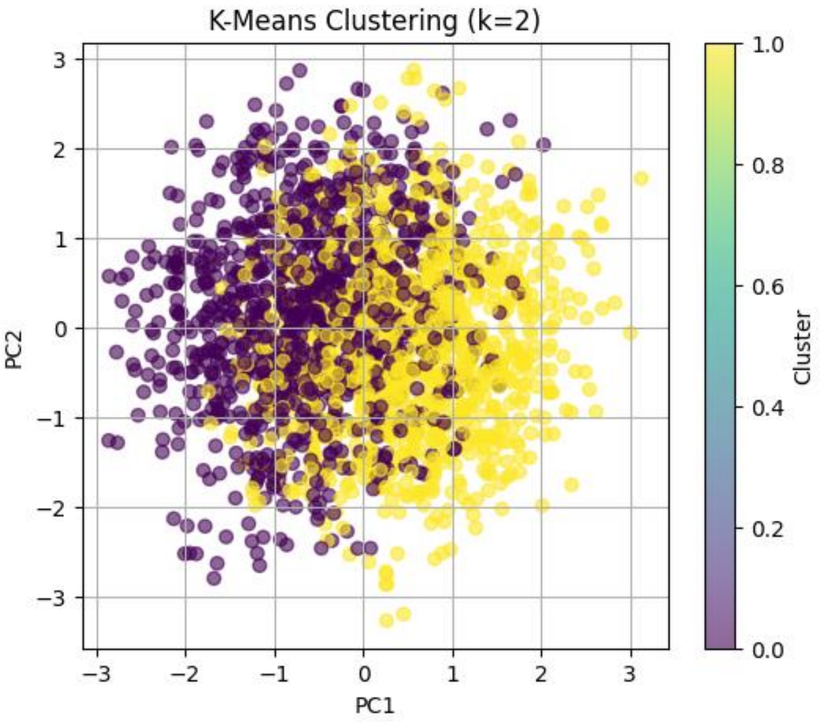
K-Means Clustering (k = 2) and Silhouette Score

After choosing 2 clusters, we applied K-Means to the scaled dataset. The PCA scatter plot shows the two clusters separated visually in 2D space. Each point represents an individual and the colors represent their assigned cluster.

The models quality was evaluated using the **silhouette score**, which measures how well-separated the clusters are.

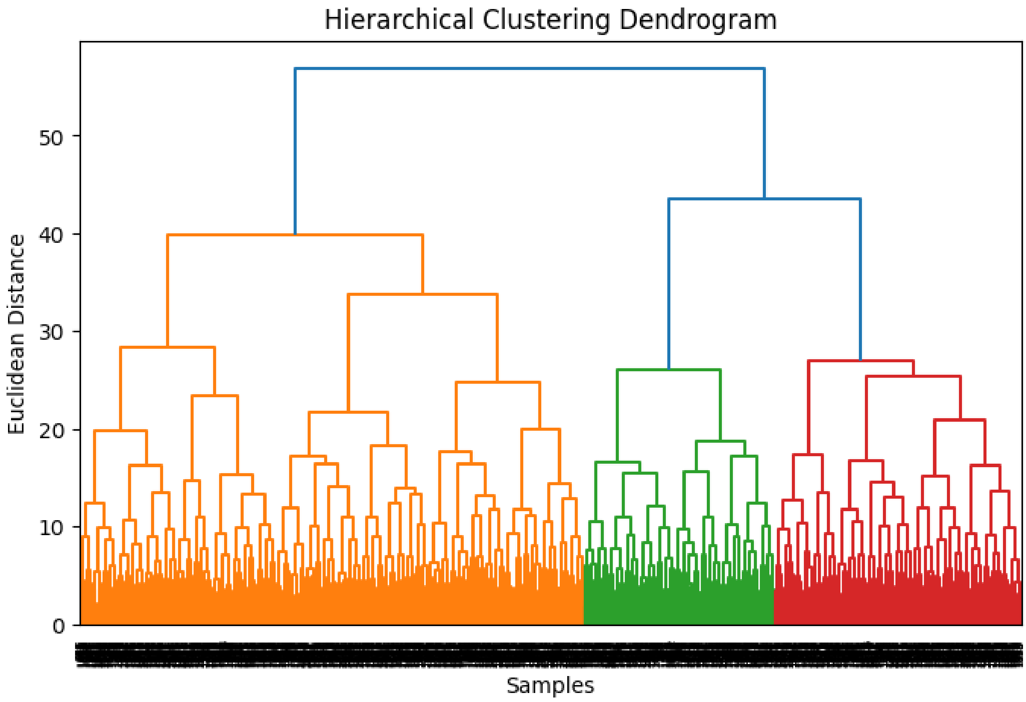
Silhouette Score: 0.102

A higher positive score indicates clearer, better-defined clusters. For this dataset, the silhouette score shows the clustering structure is **moderately separated**, meaning K-Means with 2 clusters provides a reasonable grouping.



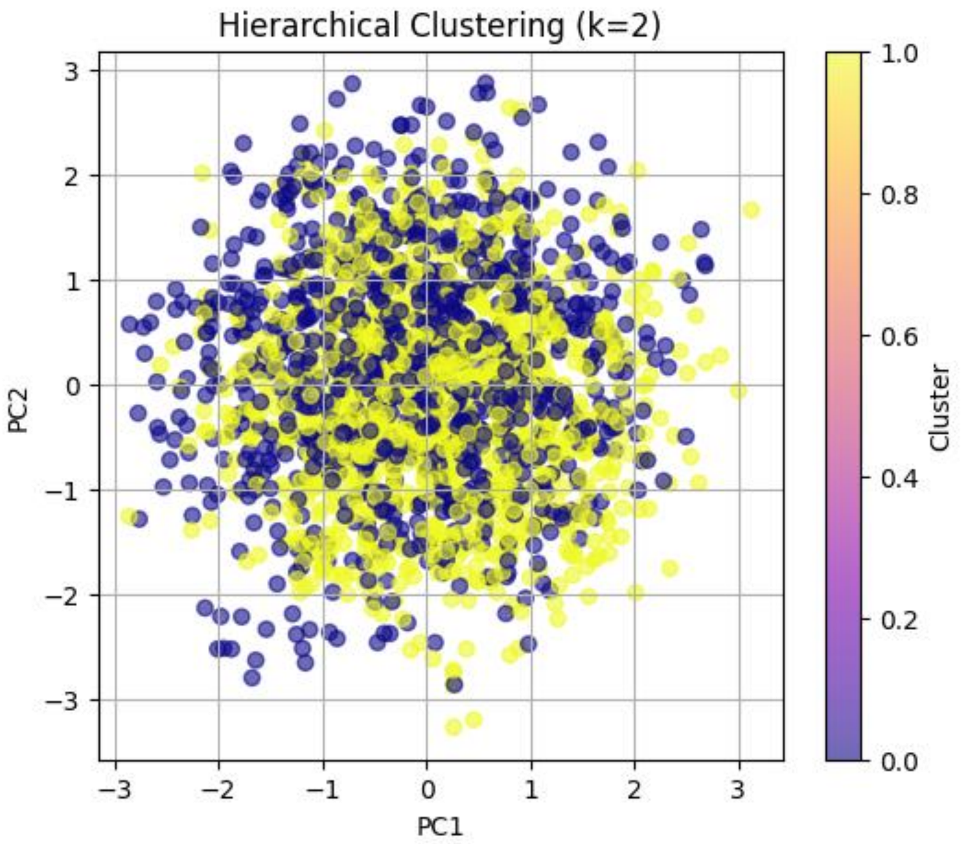
Visualization: Hierarchical Clustering Dendrogram

The dendrogram visualizes how data points merge together step-by-step using hierarchical clustering. It shows the distances at which clusters combine helping identify a suitable number of clusters. In this plot, two major branches emerge, supporting the choice of **k = 2** as the dataset naturally separates into two broad groups.



Hierarchical Clustering (k = 2)

We applied Agglomerative Hierarchical Clustering using Euclidean distance and Ward's linkage method with 2 clusters. The PCA visualization shows how the two clusters are distributed in the 2D space. Unlike K-Means the clusters appear more mixed, indicating that the data does not form very strong natural groups.



Cluster Summary – Simple Conclusion

Both K-Means and Hierarchical Clustering created **2 clusters**, but the results show that:

- The averages of features like **Age, Work Pressure, Job Satisfaction, Work Hours and Financial Stress** are almost the same in both clusters.
- There is **no big difference** between the groups.

Final Conclusion

The data does **not form strong or clear clusters**.

Both methods show that the people in this dataset are **very similar**, so clustering does not separate them well.

Conclusion output:

```
K-Means Cluster Summary (numeric features):
                      Age  Work Pressure  Job Satisfaction  Work Hours  \
KMeans_Cluster
0               42.292659       2.955357          2.996032    6.045635
1               42.055449       3.086042          3.033461    5.820268

                Financial Stress  Hier_Cluster
KMeans_Cluster
0                       2.935516      0.484127
1                       3.020076      0.586998

Hierarchical Cluster Summary (numeric features):
                    Age  Work Pressure  Job Satisfaction  Work Hours  \
Hier_Cluster
0             42.392857       3.097689          3.007353    5.789916
1             41.980944       2.956443          3.021779    6.052632

              Financial Stress  KMeans_Cluster
Hier_Cluster
0                     3.003151        0.453782
1                     2.957350        0.557169
```

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